

him.AI.Tech

High-precision Intelligent Metrology AI Technology

Autonomous 4K JIG Inspection & Metrology Total Solution

INTELLIGENT INSPECTION REVOLUTION

Harnessing the power of 4K ultra-resolution and AI to bridge the gap between physical manufacturing and digital blueprints.

TOTAL SOLUTION ECOSYSTEM



STEP-to-Inspect

Zero-teaching workflow. Upload any STEP file and let the AI generate inspection paths and 4K mapping instantly.



4K Vision Synergy

Integration of ultra-resolution 4K vision and high-speed linear motor rails for autonomous best-view finding.



Autonomous Verdict

Real-time DB matching and 100% automated Pass/Fail decisions based on high-fidelity 4K imaging.

HARDWARE MASTERY

Ultra-Resolution 4K Vision

Our 4K cameras capture detail far beyond standard industrial sensors. Integrated with linear motor rails, the system scans the JIG at 4K resolution to find the optimal perspective.

This dynamic 4K positioning ensures sub-micron data acquisition for every primitive, regardless of the part's complexity.

ial Camera here

025 guide helps U.S.
industrial camera
AI-driven QC.

AUTONOMOUS CAD PROCESSING



Plug-and-Play CAD

Simply upload your STEP file. him.AI.Tech's engine parses multi-layer primitive data and prepares the 4K scan map instantly.

4K Data Alignment

The AI correlates CAD geometry with 4K camera frames, identifying Holes, Pockets, and Slots without manual teaching.

PRECISION & SPEED IMPACT

0.01

Manufacturing Tolerance (mm)

1.0

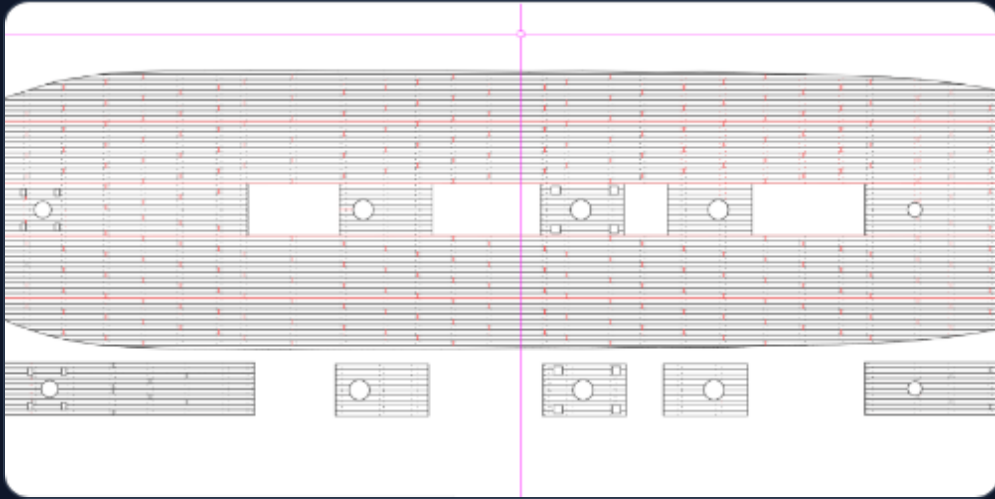
4K Vision Detection Limit (mm)

1

Total Scan Time (min)

Transforming traditional 24-hour manual inspection into a high-fidelity 60-second 4K autonomous routine.

INTELLIGENT VIEW ANALYSIS



Top 4K Perspective

UHD detail captures primary surface alignment and complex pocket patterns for global matching.



Bottom 4K Perspective

AI classifies distinct cavity patterns from the bottom view to retrieve optimized 4K DB primitives.

him.AI.Tech handles complex 4K perspectives in real-time, matching local primitives regardless of orientation.

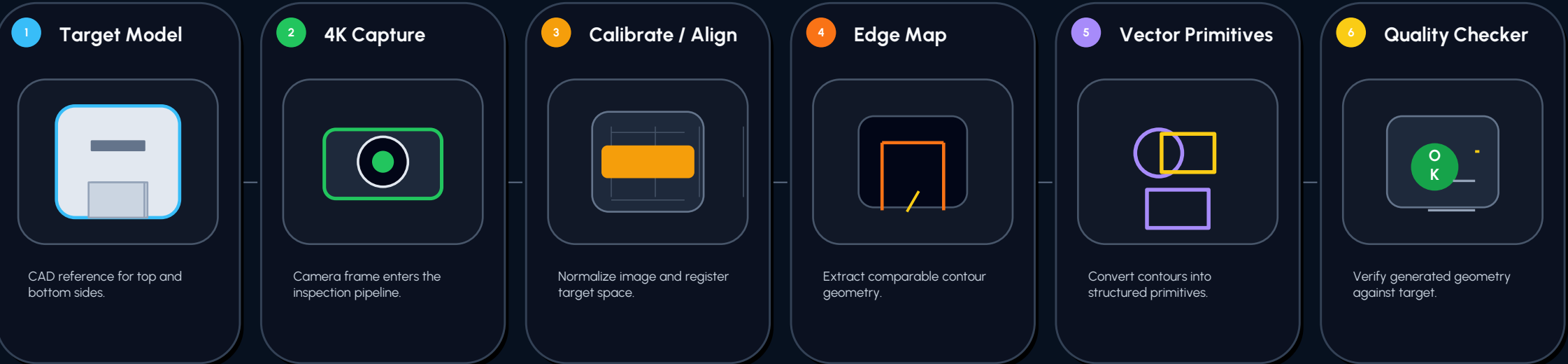
hima AI.Tech Technology Overview Flow

Generated concept view | same pipeline logic without real image dependency

CONCEPT ARCHITECTURE

Conceptual Metrology Architecture

Generated schematic cards preserve the same sequence: target, capture, calibration, edge extraction, vectorization, and checking.



Operating Principle

Separate top and bottom lanes produce aligned geometry, vector primitives, and checker-ready quality evidence.

Top lane

Bottom lane

Overlay compare

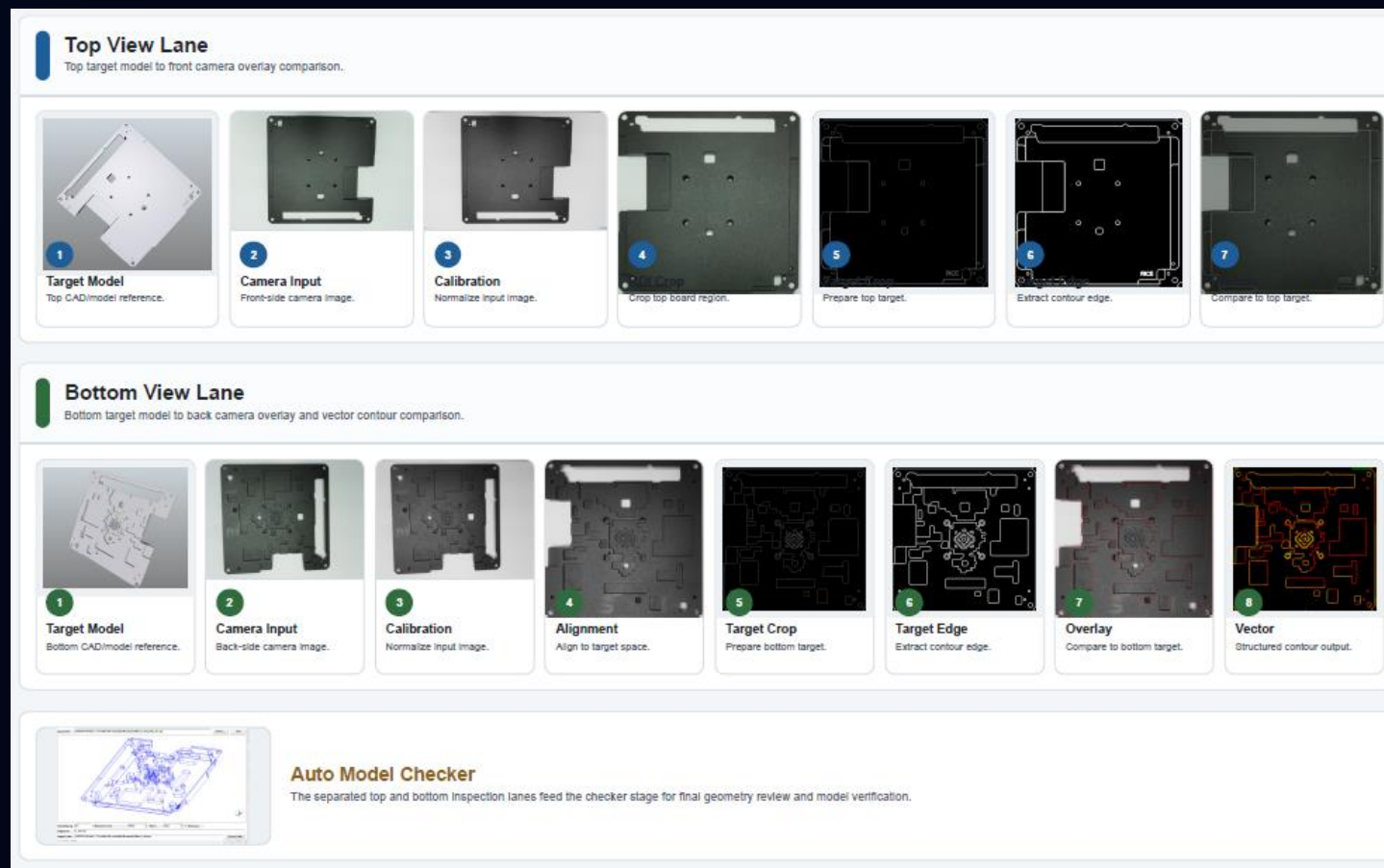
Vector data

Checker result

GUIDED MATCHING ENGINE

4K Knowledge-Based Search

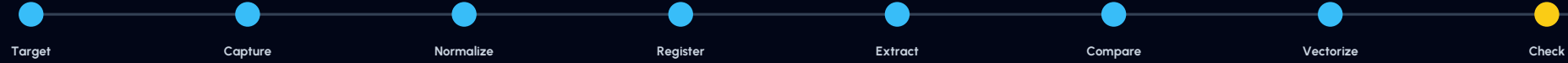
The system uses CAD data to guide the 4K vision sensor, constraining the search space for sub-pixel accuracy. Feature extraction is handled at the sub-pixel level in 4K resolution. This allows the system to convert image coordinates into absolute mm-units for direct, high-confidence metrological verification.



hima AI.Tech Technology Overview Flow

Actual video/image evidence | separated top and bottom metrology lanes

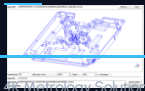
EVIDENCE PIPELINE



TOP VIEW LANE



BOTTOM VIEW LANE



Auto Model Checker

Top and bottom lanes stay separate through visual processing, then feed final geometry review and model verification.

FINAL QA

4K Visual Integrity Proof

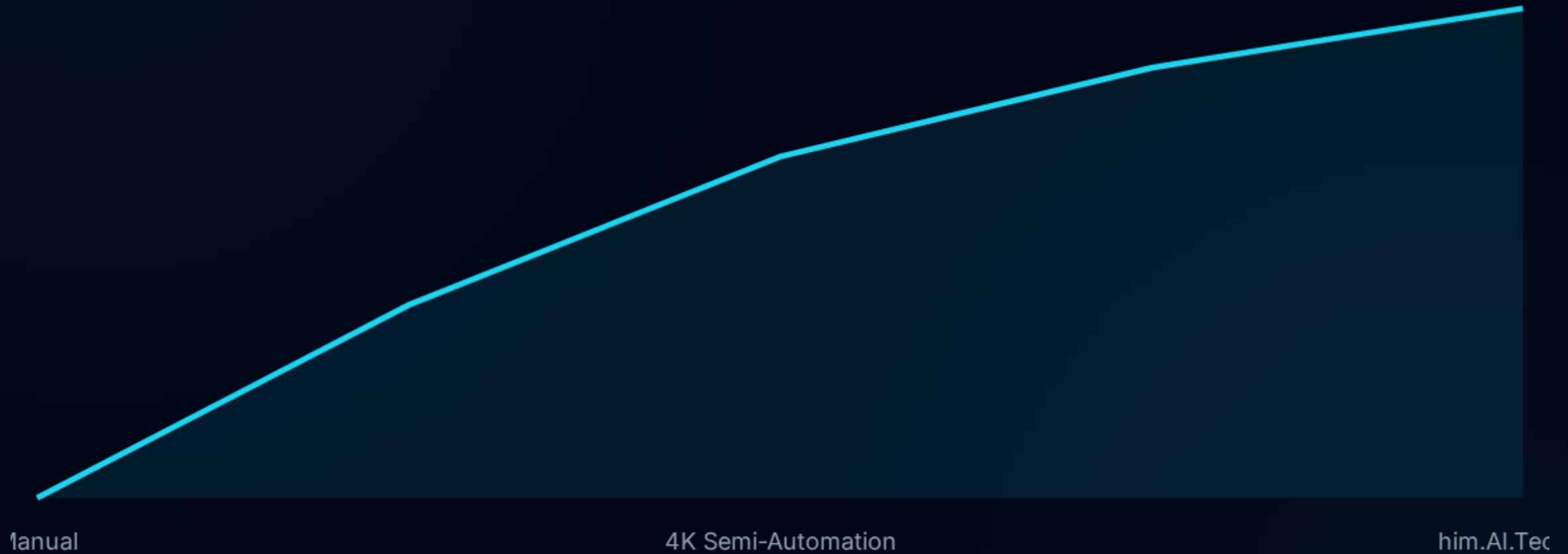
Our system overlays the red CAD wireframe directly onto the 4K frame, providing an undeniable proof of mm-scale fidelity for the manufactured JIG.

COMPARATIVE EXCELLENCE

Inspection Metric	Standard Manual Process	him.AI.Tech 4K Solution
Setup Time	3 - 5 Hours (Manual Teaching)	< 5 Minutes (STEP Upload)
Resolution	Limited to Human Vision	Ultra-High 4K Imaging
Scan Speed	Variable (Human Dependent)	< 60 Seconds (Autonomous)
Precision Level	Subjective Visuals	±1.0 mm (Automated Metrology)

LEAD TIME OPTIMIZATION

LEAD-TIME OPTIMIZATION



Exponential lead time reduction driven by ultra-resolution 4K intelligence.

Questions?

Lead the future of manufacturing with him.AI.Tech 4K Metrology.

www.him-ai-tech.com
contact@him-ai-tech.com

IMAGE SOURCES



[https://cdn.prod.website-files.com/67e4e66460f2e4ed3eae3aca/691aaf99efb6ece90ebef4fb_\(20\)%205%20Best%20Industrial%20Camera%20Systems%20Out%20There.jpg](https://cdn.prod.website-files.com/67e4e66460f2e4ed3eae3aca/691aaf99efb6ece90ebef4fb_(20)%205%20Best%20Industrial%20Camera%20Systems%20Out%20There.jpg)

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Source: www.baileigh.com



<https://honeywell.scene7.com/is/image/honeywell/hon-aero-digital-machining-wireframe-overlay:1-1-square?wid=1245&hei=1245&dpr=off>

Source: aerospace.honeywell.com